

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

PREPX

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**PREPX**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Rajat Takkar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Kalash Thakur

Date: 06-05-26

2211985023

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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I would like to express my sincere gratitude to my project guide, Dr. Rajat Takkar, for his constant guidance, valuable insights, and unwavering support throughout the development of my project, *PREPX*. His expertise and encouragement played a crucial role in shaping this work and enhancing my understanding of the subject. I am also thankful to Chitkara University Rajpura for providing a conducive learning environment and the necessary resources to successfully complete this project. Furthermore, I am grateful to my peers and teammates for their collaboration, idea-sharing, and continuous engagement, which contributed to refining the overall quality of the project. The collective effort and discussions played a vital role in overcoming challenges and achieving the desired outcomes.

ABSTRACT

In today's competitive job market, students often struggle to prepare effectively for campus placements. They face challenges due to inadequate guidance, limited exposure to real interview situations, and a lack of understanding about industry expectations. Many students depend on scattered online resources that do not match their skill levels. This leads to inefficient preparation and lower confidence during recruitment. Moreover, educational institutions typically do not have effective tools to track student readiness, identify performance gaps, and offer targeted support based on individual learning needs. The goal of this project is to create a platform to help students prepare for interviews. This platform will offer structured learning, mock assessments, and intelligent feedback. Key features include secure user authentication and personalized learning recommendations. The platform will help students practice interview questions and improve their performance through automated evaluation. It will incorporate secure user authentication to ensure data privacy and manage access based on user roles. The system will also provide feedback on student responses. The system will be developed using the MERN stack, which consists of React for building a responsive and interactive frontend, Node.js with Express for managing backend services and APIs, and MongoDB for efficient data storage and management. The platform will also feature a comprehensive analytics dashboard that provides actionable insights for both students and faculty. Students will be able to visualize their progress over time, track scores across different categories, and set personal goals. Security and scalability are central to the platform's architecture. All user data will be protected through industry-standard encryption protocols, and JWT-based authentication will ensure that access is strictly controlled based on user roles. The application will be built to scale horizontally, accommodating a growing number of concurrent users during peak placement seasons without compromising performance or reliability. Ultimately, this platform aims to bridge the gap between academic learning and industry expectations by creating a structured, data-driven, and student-centric preparation ecosystem. By combining intelligent automation with personalized guidance, it seeks to empower every student with the tools, knowledge, and confidence needed to succeed in campus placements and beyond.

Introduction

The rapid growth of the technology industry has created an ever-increasing demand for skilled professionals, making campus placements one of the most critical milestones in a student's academic journey. However, the path to securing a desirable placement is far from straightforward. Despite possessing theoretical knowledge gained through years of formal education, many students find themselves underprepared when confronted with the practical demands of technical interviews, aptitude assessments, and behavioural evaluations conducted by recruiters. There is, therefore, a pressing need for a comprehensive, intelligent, and institutionally integrated platform that not only equips students with the right knowledge and practice but also enables educators to monitor progress, identify gaps, and provide timely intervention. This project proposes the development of such a platform, built using the MERN stack, that combines structured learning, automated assessments, personalized feedback, and insightful analytics to transform the way students prepare for campus placements and step into their professional careers.

Problem Background:

The transition from academic life to professional employment represents one of the most challenging phases in a student's career. Campus placement drives serve as the primary gateway through which students secure their first employment opportunities, yet a significant proportion of students continue to underperform during these recruitment processes despite completing their academic programs. This disconnect between academic achievement and placement readiness has become a growing concern for students, educators, and institutions across the country. Another critical dimension of the problem lies within educational institutions themselves. Most colleges and universities do not possess adequate tools or systems to effectively monitor the placement preparedness of their students at an individual level. Faculty members and placement coordinators are often unable to identify which students are struggling, in which specific areas they are falling short, and what kind of targeted support would be most beneficial for them.

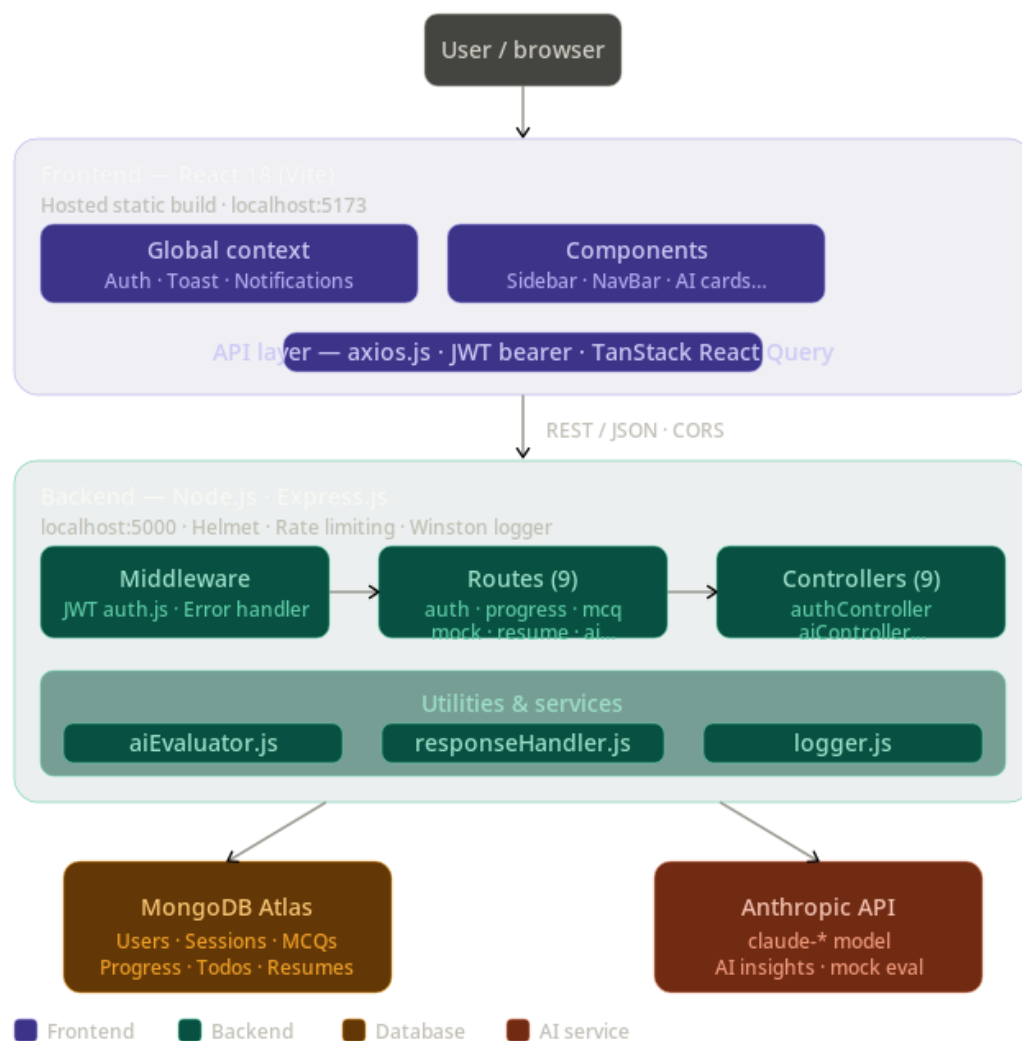
Objective of the project:

The primary objective of this project is to design and develop a comprehensive, intelligent, and user-centric interview preparation platform that addresses the existing gaps in campus placement readiness among students. The platform aims to provide a structured, personalized, and technology-driven learning environment that bridges the divide between academic knowledge and industry expectations. The platform further aims to consolidate interview

preparation resources into a single, well-organized repository that students can access at any point in their preparation journey.

Proposed solution / System overview:

To address the multifaceted challenges associated with campus placement preparation, this project proposes the development of an intelligent, web-based interview preparation platform built on the MERN stack. The proposed solution is designed to serve as a unified, end-to-end ecosystem that consolidates structured learning, automated assessments, personalized feedback, and institutional analytics into a single, accessible, and scalable application. The platform is envisioned as a technology-driven bridge that connects students with the knowledge, practice, and guidance they need to confidently navigate the campus recruitment process.



Key Features or Functionality

The interview preparation platform is designed to deliver a rich and comprehensive set of features that collectively address every dimension of the campus placement preparation process. At the foundation of the platform lies a secure and role-based authentication system that governs user registration, login, and access control. Each user, whether a student or an administrator, is assigned a specific role that determines the features and data they can access, ensuring that the platform remains organized, secure, and appropriately personalized for every category of user.

1. Structured Learning Paths

Prepx offers well-organized preparation tracks for different domains like technical, HR, and domain-specific interviews.

- Curriculum aligned with real interview patterns
- Step-by-step progression from basics to advanced

2. Question Bank & Practice Modules

A comprehensive repository of curated questions

- Previous year interview questions from top companies
- Topic-wise and difficulty-wise categorization
- Timed practice sets for real exam simulation

3. Mock Tests & Assessment Engine

Designed to replicate real interview and test environments.

- Full-length mock tests with timer
- Instant score calculation with detailed breakdown

4. Company-Specific Preparation

Focused preparation modules for target companies.

- Company-wise question sets
- Interview experiences shared by previous candidates
- Hiring pattern analysis (based on historical data)

5. Resume & Profile Building Tools

Helps users present themselves better during interviews.

- Resume templates (ATS-friendly)
-

Methodology

1. Development Approach — Agile (Iterative)

PrepX was built using an iterative development approach, where features were planned, built, tested, and refined in short cycles. This allowed the team to progressively add modules (auth → dashboard → MCQ → mock interview → AI features) without waiting for the entire system to be complete.

2. System Design

The architecture follows a client-server separation pattern:

- Frontend and backend are completely decoupled, communicating only via REST APIs. This makes each side independently deployable and testable.
- The backend itself follows an MVC-inspired layered structure — routes handle incoming requests, controllers contain business logic, and Mongoose models define data schemas.
- Global state on the frontend is managed through React Context (Auth, Toast, Notifications), keeping component trees clean and avoiding prop drilling.

3. Technology Selection Rationale

Each technology was chosen for a specific reason:

- React + Vite — Fast development server, hot module replacement, and a component-driven UI model suited for a feature-rich SPA.
- Node.js + Express — Lightweight, JavaScript-native backend that pairs naturally with the React frontend and MongoDB's JSON document model.
- MongoDB (Mongoose) — Schema-flexible document store ideal for varied data types like resumes, MCQ sessions, and user progress records.
- JWT Authentication — Stateless auth that scales well without server-side session storage; tokens are sent as bearer headers on every protected API call.
- Anthropic API — Provides the AI evaluation layer for mock interviews and answer scoring without requiring a custom ML model.

4. Security Implementation

Security was treated as a cross-cutting concern, not an afterthought:

- Helmet.js sets HTTP security headers on every response.
- Rate limiting is applied globally (100 req / 15 min) and more aggressively on sensitive routes like /api/auth and /api/ai (20 req / hour).
- bcryptjs hashes passwords before storage — plaintext credentials are never persisted.
- JWT middleware (auth.js) guards all protected routes, rejecting requests without a valid token.

- CORS is explicitly configured to accept requests only from the known frontend origin.

5. AI Integration Strategy

AI features are integrated as an optional enrichment layer rather than a core dependency:

- The `aiEvaluator.js` utility wraps Anthropic API calls, keeping AI logic isolated from core controllers.
- The `ANTHROPIC_API_KEY` is optional in `.env` — the app runs without it, degrading gracefully on AI features.
- Rate limiting on `/api/ai` protects against excessive API usage and cost overruns.

6. Data Modeling

Six Mongoose models cover all persistence needs: `User`, `MCQ`, `MockInterviewSession`, `Progress`, `Resume`, and `Todo`. Each model maps to a distinct feature domain, keeping schemas focused and avoiding over-normalization that would complicate queries.

7. Logging & Observability

Winston logger captures all requests and errors to `combined.log` and `error.log`. The `responseHandler.js` utility standardizes all API responses, ensuring consistent JSON structure (success, message, data) across every endpoint — making frontend error handling predictable.

8. Frontend UX Strategy

- Framer Motion provides page transition animations, giving the app a polished, app-like feel.
- Three.js renders a 3D background on auth pages to create a distinctive visual identity.
- TanStack React Query manages server-state caching, reducing redundant API calls and keeping the UI snappy.
- `ProtectedRoute` wraps all authenticated pages, redirecting unauthenticated users to `/login` without exposing any page content.
-

9. Testing & Validation

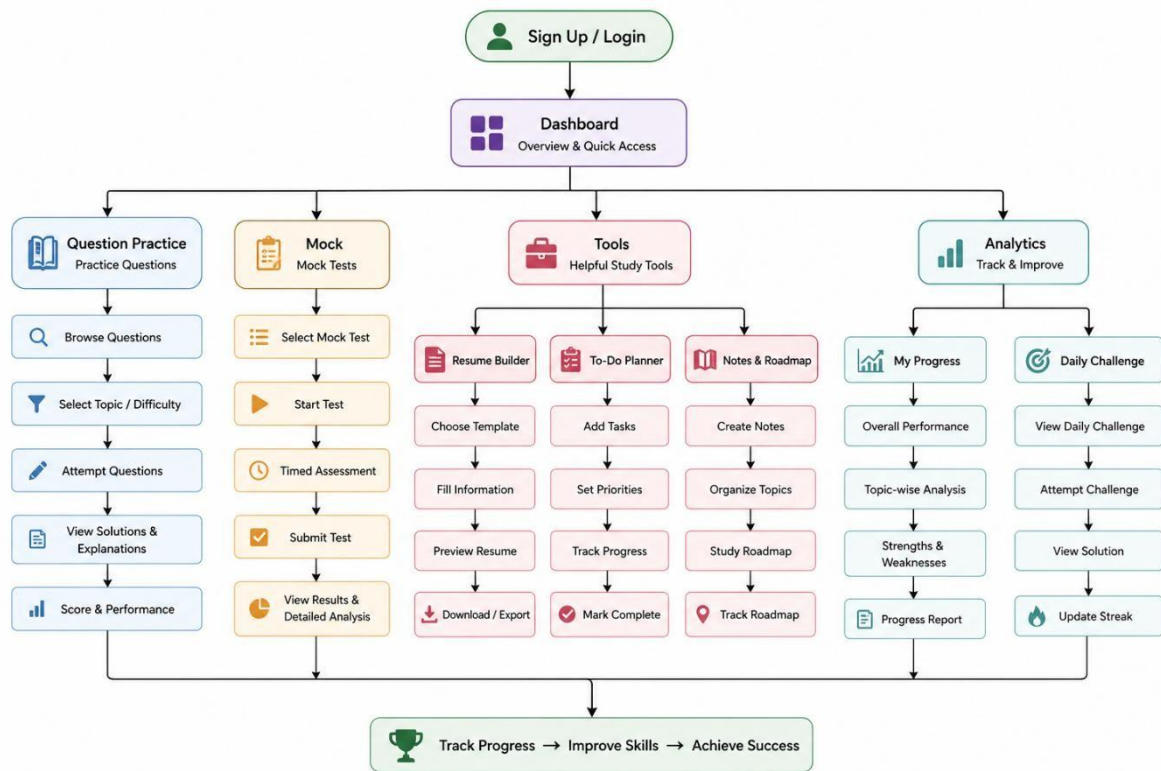
The project relied on manual integration testing across feature modules, with the backend health check endpoint (`GET /api/health`) serving as a quick liveness probe during development.

10. Future Roadmap

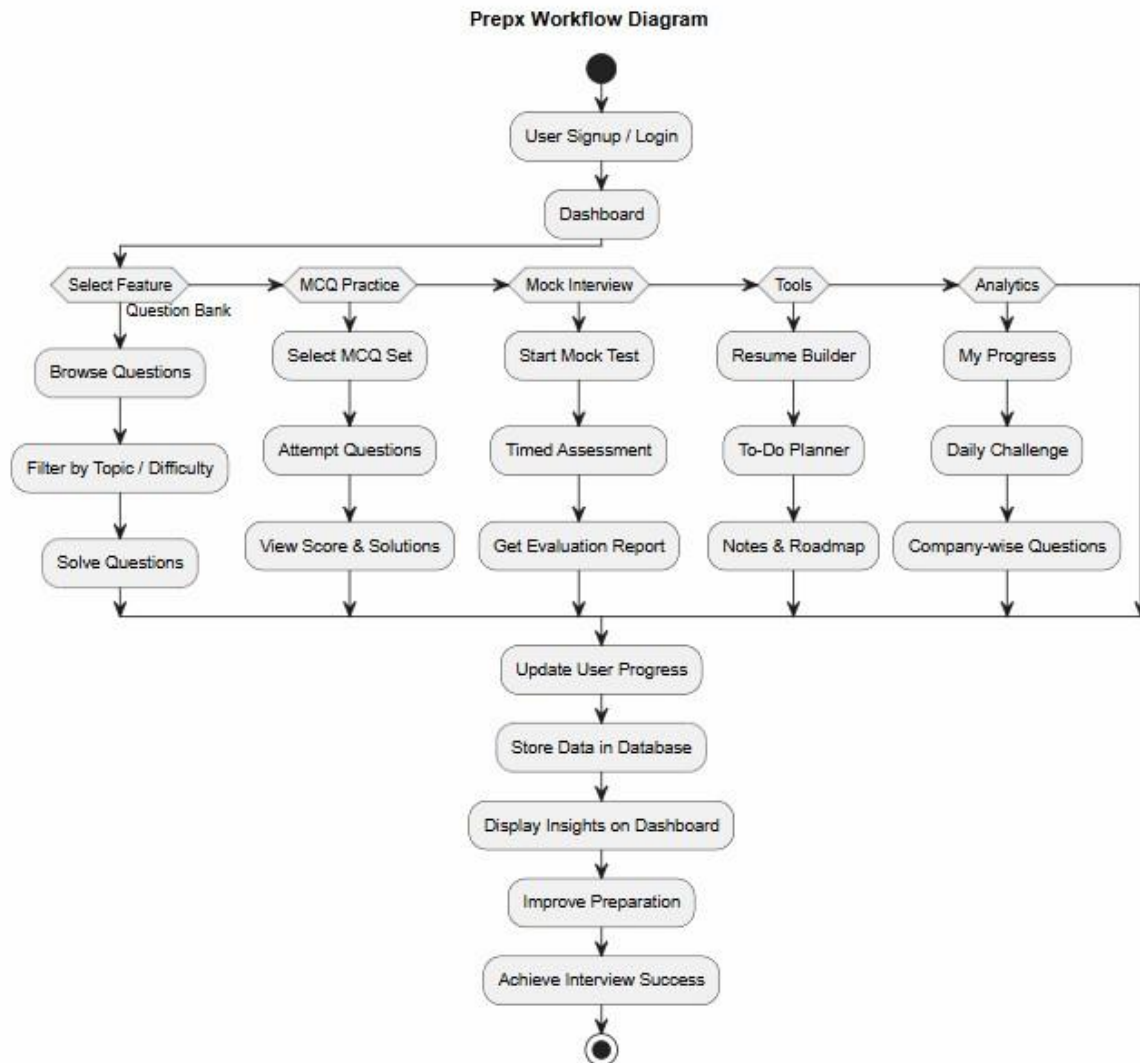
The team identified clear next steps: a voice-based AI interview assistant, a leaderboard system, email reminders, personalized study roadmaps, and company-specific preparation paths — all of which fit naturally into the existing modular architecture without requiring structural changes.

Diagrams

Flow chart:



Workflow Diagram:



1. Authentication and Entry

- The user begins by Sign Up / Login into the system.
- After successful authentication, the user is redirected to the Dashboard, which acts as the central control panel for all activities.

2. Feature Selection and Navigation

- From the dashboard, the user selects a feature such as Question Bank, MCQ Practice, Mock Interview, Tools, or Analytics.
- Each feature represents a different preparation pathway, allowing the user to customize their learning journey.

3. Practice and Learning Execution

- **Question Bank Flow**

- The user browses available questions.
- Questions are filtered by topic or difficulty level.
- The user solves the selected questions to build conceptual understanding.

- **MCQ Practice Flow**

- The user selects an MCQ set.
- Attempts objective questions in a structured format.
- Views scores and detailed solutions for performance improvement.

- **Mock Interview Flow**

- The user starts a mock test.
- Completes a timed assessment simulating real interview condition.
- Receives an evaluation report with performance insights.

4. Productivity Tools Utilization

- The user accesses tools like Resume Builder, To-Do Planner, and Notes & Roadmap.
- Resume Builder helps in creating professional resumes.
- To-Do Planner organizes daily preparation tasks.
- Notes & Roadmap supports structured learning and revision planning.

5. Analytics and Performance Tracking

- The user checks My Progress to monitor overall performance.
- Participates in Daily Challenges to maintain consistency.
- Practices Company-wise Questions for targeted preparation.

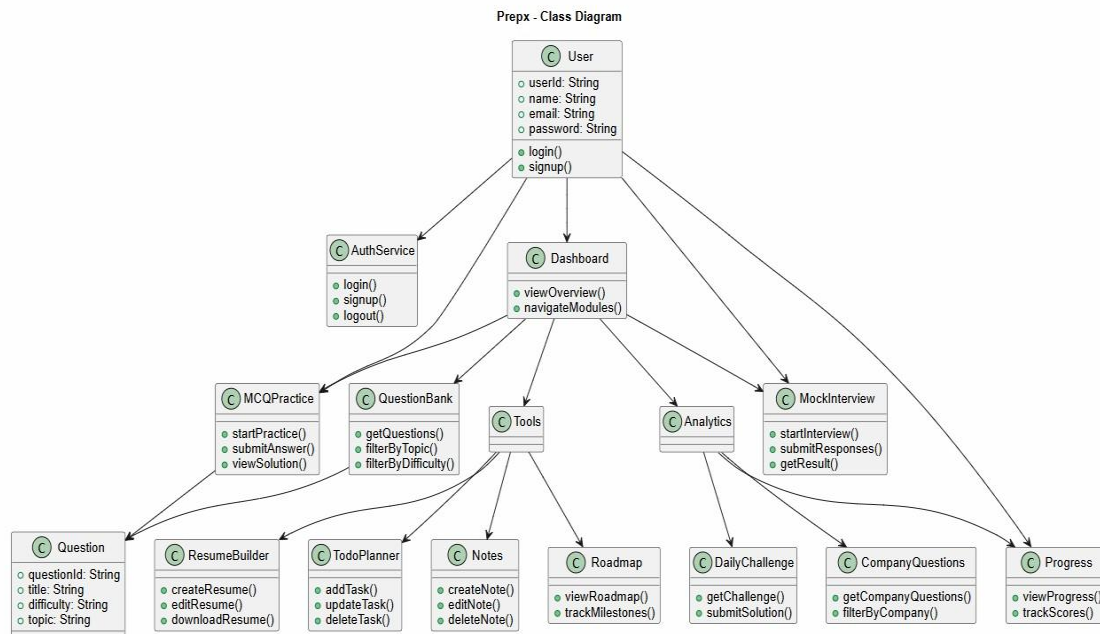
6. Data Processing and Storage

- After completing activities, the system updates user progress.
- All performance data, attempts, and results are stored in the database (MongoDB).
- The backend processes this data to maintain accuracy and consistency.

7. Insights and Continuous Improvement

- The system displays insights on the dashboard, including scores, accuracy, and trends.
- Based on these insights, the user identifies strengths and weaknesses.
- The user refines their preparation strategy accordingly.

Class Diagram:



1. Core Objective

The goal of Prepx is to provide a structured, end-to-end interview preparation platform where users can practice, track progress, and improve consistently. Instead of random preparation, the system creates a modular learning ecosystem—combining question practice, mock interviews, productivity tools, and analytics into one unified workflow. The architecture ensures that every user action contributes to measurable growth and interview readiness.

2. The Technical Backbone (MERN Architecture)

The platform is designed using a modular class-based structure mapped to the MERN stack, ensuring scalability and maintainability:

- **Frontend Layer:** Handles user interaction through dashboards and modules (mapped to classes like *Dashboard*, *Tools*, *Analytics*).
- **Backend Layer:** Managed using service-based logic (e.g., *AuthService*) to process authentication and workflows.

- **Database Layer:** Stores structured data like users, questions, progress, and tasks (represented by classes such as *User*, *Question*, *Progress*).

Each class represents a functional unit, ensuring separation of concerns and smooth data handling across the system.

3. How the System Works (Class Interactions)

• User & Authentication Layer

- The User class is the central entity containing attributes like `userId`, `name`, `email`, and `password`.
- It interacts with `AuthService`, which manages login, signup, and logout operations. • This ensures secure and controlled access to the platform

• Dashboard as Control Hub

- The Dashboard class acts as the orchestrator, allowing users to view an overview and navigate between modules.
- It connects all major components like `QuestionBank`, `MCQPractice`, `MockInterview`, `Tools`, and `Analytics`.

• Practice & Learning Modules

- **QuestionBank:** Retrieves and filters questions by topic and difficulty.
- **MCQPractice:** Handles practice sessions, answer submission, and solution viewing.
- **MockInterview:** Simulates interview environments with test execution and result generation.

All these modules rely on the `Question` class, which stores core question data (`ID`, `title`, `difficulty`, `topic`).

• Productivity & Support Tools

- **Tools** acts as a parent module connecting:
- **ResumeBuilder:** Create, edit, and download resumes

- **ToDoPlanner:** Manage daily tasks
- **Notes:** Create and manage notes
- **Roadmap:** Track milestones and preparation path

This layer ensures users can plan, organize, and optimize their preparation strategy.

• **Analytics & Performance Tracking**

- **Analytics** connects to:
 - **DailyChallenge:** Provides daily practice tasks
 - **CompanyQuestions:** Offers company-specific preparation
 - **Progress:** Tracks scores and overall performance
- The Progress class stores and manages user performance metrics, enabling data-driven improvement.

4. Summary of the Architecture

In essence, Prepx is not just a practice platform—it is a modular preparation engine. The system is built around a central user entity, connected to multiple independent yet integrated modules.

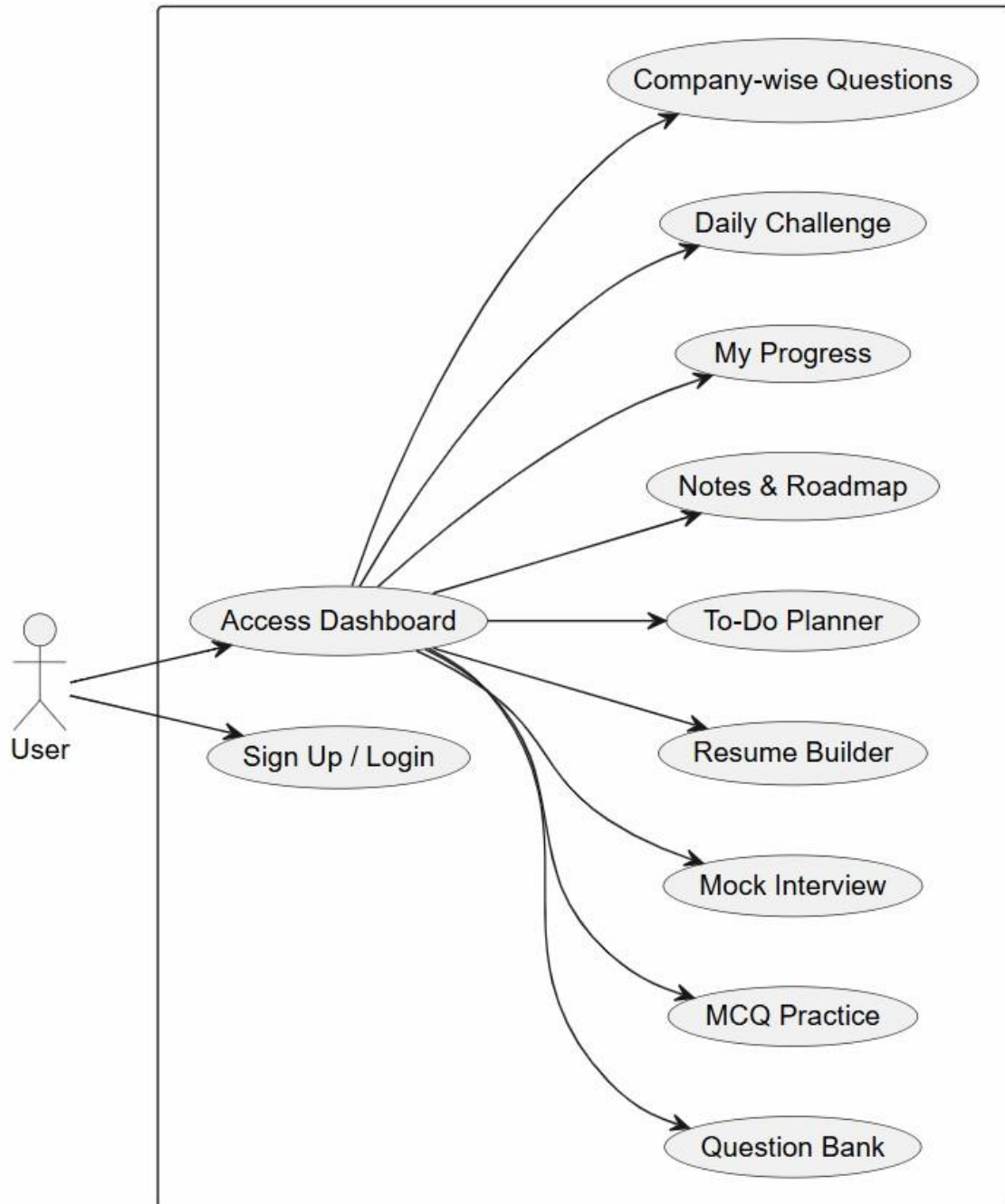
- The Dashboard acts as the control layer
- Practice modules build knowledge and skills
- Tools enhance productivity and planning
- Analytics provides continuous feedback and improvement

This architecture ensures:

- **Scalability:** New features can be added easily
- **Maintainability:** Each module works independently
- **Efficiency:** Smooth data flow between components

Use case diagram :

Prepx - Use Case Diagram



1. The Actor

- **User:** The primary actor of the system. This represents any student or candidate using Prepx for interview preparation. The user interacts with all core features through the platform.

2. Functional Workflows by Actor

• Authentication Flow

- The user begins with Sign Up / Login, which acts as the entry point into the system.
- After successful authentication, the user moves to the Access Dashboard, which serves as the central control hub.

• Dashboard-Based Actions

Once inside the dashboard, the user can navigate multiple modules:

Core Preparation Modules

- **Question Bank:** Browse structured interview questions by topic or company.
- **MCQ Practice:** Attempt objective-based questions for quick revision and accuracy building.
- **Mock Interview:** Simulate real interview scenarios to improve confidence and performance.

Productivity & Support Tools

- **Resume Builder:** Create and manage professional resumes.
- **To-Do Planner:** Organize daily preparation tasks and goals.
- **Notes & Roadmap:** Maintain study notes and follow a structured preparation path.

Analytics & Growth Tracking

- **My Progress:** Track performance metrics like scores, accuracy, and improvement trends.
- **Daily Challenge:** Engage with daily tasks to maintain consistency.

- **Company-wise Questions:** Prepare specifically for target companies based on past questions and patterns.

- **Relationship Insight**

- The Access Dashboard acts as a central node, connecting all functionalities.
- All features are indirectly dependent on authentication, meaning the user must log in before accessing any module.

3. Shared Capabilities

- **Centralized Access:** All modules are accessible through a single dashboard, ensuring a smooth user experience.
- **Integrated Learning Flow:** Practice, testing, planning, and tracking are interconnected to provide end-to-end preparation.
- **Continuous Feedback Loop:** Features like progress tracking and daily challenges help users continuously improve

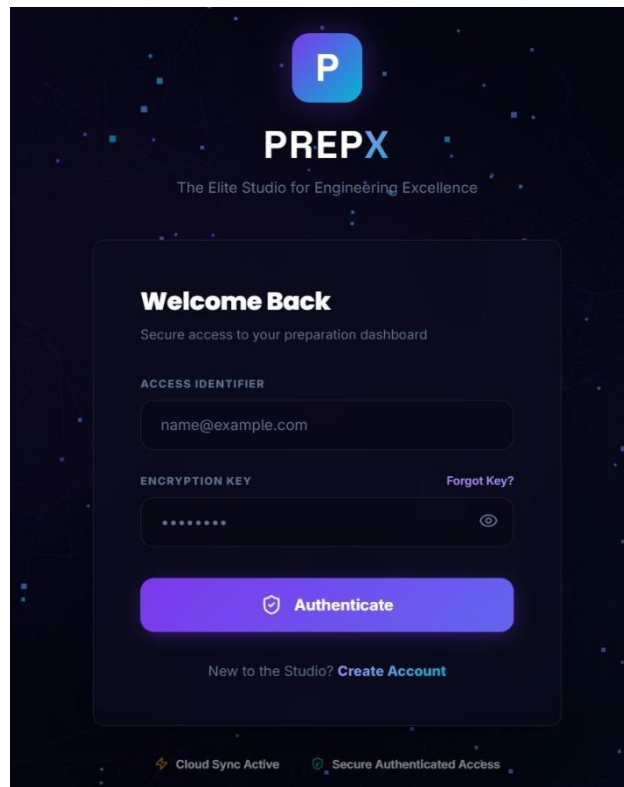
4. Technical Importance

This diagram highlights the system architecture from a user interaction perspective. The dashboard acts as a control layer, ensuring modular access to all services. This design supports:

- **Scalability:** New modules can be added without affecting existing ones
- **Maintainability:** Each feature works as an independent unit
- **User Experience Optimization:** Single entry point reduces complexity

Snapshots:

Login



The login interface for PREP X features a dark blue background with a subtle geometric pattern. At the top center is a logo consisting of a blue square with a white 'P' inside, followed by the text 'PREP X' in a bold, white, sans-serif font. Below this is the tagline 'The Elite Studio for Engineering Excellence' in a smaller, lighter font. The main content area is a dark blue rounded rectangle with a white border. It contains the heading 'Welcome Back' in bold white text, followed by the subtitle 'Secure access to your preparation dashboard'. Below this are two input fields: 'ACCESS IDENTIFIER' with the placeholder 'name@example.com' and 'ENCRYPTION KEY' with a masked password '*****'. A 'Forgot Key?' link is positioned to the right of the encryption key field. A large, rounded purple button with a white shield icon and the text 'Authenticate' is centered below the input fields. At the bottom of the form, a link 'New to the Studio? Create Account' is displayed. At the very bottom of the interface, two status indicators are shown: 'Cloud Sync Active' with a lightning bolt icon and 'Secure Authenticated Access' with a checkmark icon.

PREP X
The Elite Studio for Engineering Excellence

Welcome Back
Secure access to your preparation dashboard

ACCESS IDENTIFIER
name@example.com

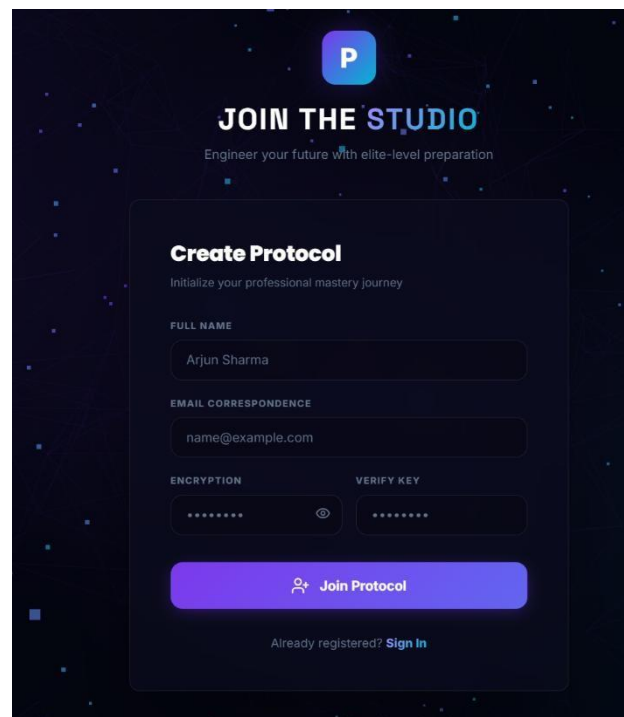
ENCRYPTION KEY
***** [Forgot Key?](#)

Authenticate

New to the Studio? [Create Account](#)

Cloud Sync Active Secure Authenticated Access

Signup



The signup interface for PREP X features a dark blue background with a subtle geometric pattern. At the top center is a logo consisting of a blue square with a white 'P' inside, followed by the text 'JOIN THE STUDIO' in a bold, white, sans-serif font. Below this is the tagline 'Engineer your future with elite-level preparation' in a smaller, lighter font. The main content area is a dark blue rounded rectangle with a white border. It contains the heading 'Create Protocol' in bold white text, followed by the subtitle 'Initialize your professional mastery journey'. Below this are four input fields: 'FULL NAME' with the placeholder 'Arjun Sharma', 'EMAIL CORRESPONDENCE' with the placeholder 'name@example.com', 'ENCRYPTION' with a masked password '*****', and 'VERIFY KEY' with a masked password '*****'. A large, rounded purple button with a white plus icon and the text 'Join Protocol' is centered below the input fields. At the bottom of the form, a link 'Already registered? Sign In' is displayed. At the very bottom of the interface, two status indicators are shown: 'End-to-End Secure' with a checkmark icon and 'Modified Engineering Tools' with a gear icon.

JOIN THE STUDIO
Engineer your future with elite-level preparation

Create Protocol
Initialize your professional mastery journey

FULL NAME
Arjun Sharma

EMAIL CORRESPONDENCE
name@example.com

ENCRYPTION

VERIFY KEY

Join Protocol

Already registered? [Sign In](#)

End-to-End Secure Modified Engineering Tools

Dashboard

The dashboard features a dark theme with a starry background. At the top, there's a search bar and a user profile for 'hamza ali madari' (Student). The main heading is 'Master Your Engineering Interviews.' Below it, a welcome message says 'Welcome back, hamza. Prepx is your high-performance studio for mastering technical interviews with precision and speed.' A progress indicator shows '84% Progress'. Two buttons, 'Explore Questions' and 'Practice Mock', are present. At the bottom, four stat cards display: '0 TOTAL SOLVED', '0d CURRENT STREAK', '#12 GLOBAL RANK', and '84% ACCURACY RATE'.

ACHIEVE TECHNICAL EXCELLENCE

Master Your Engineering Interviews.

Welcome back, **hamza**. Prepx is your high-performance studio for mastering technical interviews with precision and speed.

84% Progress

Explore Questions → Practice Mock

0 TOTAL SOLVED

0d CURRENT STREAK

#12 GLOBAL RANK

84% ACCURACY RATE

Profile

The profile page shows the user's information: 'hamza ali madari' (STUDENT), email 'dhurandhar@gmail.com', and roles 'Full Stack Developer' and 'Intermediate'. There's a 'Logout' button. Below this are four stat cards: '0 SOLVED QUESTIONS', '0d CURRENT STREAK', '0% ACCURACY', and '0 MOCK INTERVIEWS'. A navigation bar includes 'Overview', 'Progress', and 'Settings'. The 'Professional Context' section shows 'Full Stack Developer' as the target domain and 'Intermediate' as the experience tier. A 'Next Blueprint' section indicates the user is on track to master the 'Backend Architecture' track by solving 5 more Node.js questions.

hamza ali madari STUDENT

dhurandhar@gmail.com Full Stack Developer Intermediate

Logout

0 SOLVED QUESTIONS

0d CURRENT STREAK

0% ACCURACY

0 MOCK INTERVIEWS

Overview Progress Settings

Professional Context

TARGET DOMAIN
Full Stack Developer

EXPERIENCE TIER
Intermediate

Next Blueprint

You're on track to master the Backend Architecture track. Solving 5 more Node.js questions will unlock the next tier.

Question Bank

Question Bank

60 questions available · 0 solved

Q SEARCH

Find questions...

TOPIC

All 60

DSA 30

OS 7

DBMS 8

CN 7

HR 8

DIFFICULTY

All

Easy

Medium

Hard

☐ What is the time complexity of accessing an element in an array? DSA EASY

☐ How would you find the maximum subarray sum? (Kadane's Algorithm) DSA MEDIUM

☐ What is a Linked List and what are its types? DSA EASY

☐ How do you detect a cycle in a linked list? DSA MEDIUM

☐ What is the difference between BFS and DFS? DSA EASY

☐ What is a Balanced BST? Give examples. DSA MEDIUM

☐ What is the difference between memoization and tabulation? DSA MEDIUM

MCQ Practice

MCQ Assessment

Validate your conceptual depth with precision-targeted assessments.

0

TOTAL SUCCESSES

Q INTELLIGENCE SEARCH

Search parameters...

KNOWLEDGE DOMAIN

All

DSA

OS

DBMS

CN

HR

COMPLEXITY

All

Easy

Medium

Hard

☐ DSA EASY

What is the time complexity of accessing an element in an array by index?

☐ DSA MEDIUM

What is the time complexity of binary search?

☐ DSA EASY

What is the time complexity of searching in an unsorted array?

☐ DSA EASY

Which sorting algorithm is stable?

☐ DSA MEDIUM

Worst-case time complexity of Quick Sort?

☐ DSA EASY

Which principle does stack follow?

☐ DSA EASY

Mock Interview

Mock Interview Studio

Simulate high-pressure technical interviews with real-time feedback.

Session Configuration

SELECT FOCUS TOPIC

All DSA OS DBMS CN HR

QUESTION COUNT

3 Questions 5 Questions 10 Questions

2:00 / Q Self-Rating

Initialize Session History

Resume Builder

Resume Studio

Engineer your professional identity for elite technical roles.

Sync Export

IDENTITY ACADEMIC INDUSTRIAL PORTFOLIO EXPERTISE

Full Name

Arjun Sharma

Target Role

Software Engineer

Email Address

arjun@example.com

Phone Matrix

+91 98765 43210

Global Location

Cloud City, Earth

GitHub Repository

github.com/profile

LinkedIn Protocol

linkedin.com/in/profile

Professional Executive Summary

High-performance technical specialist with expertise in...

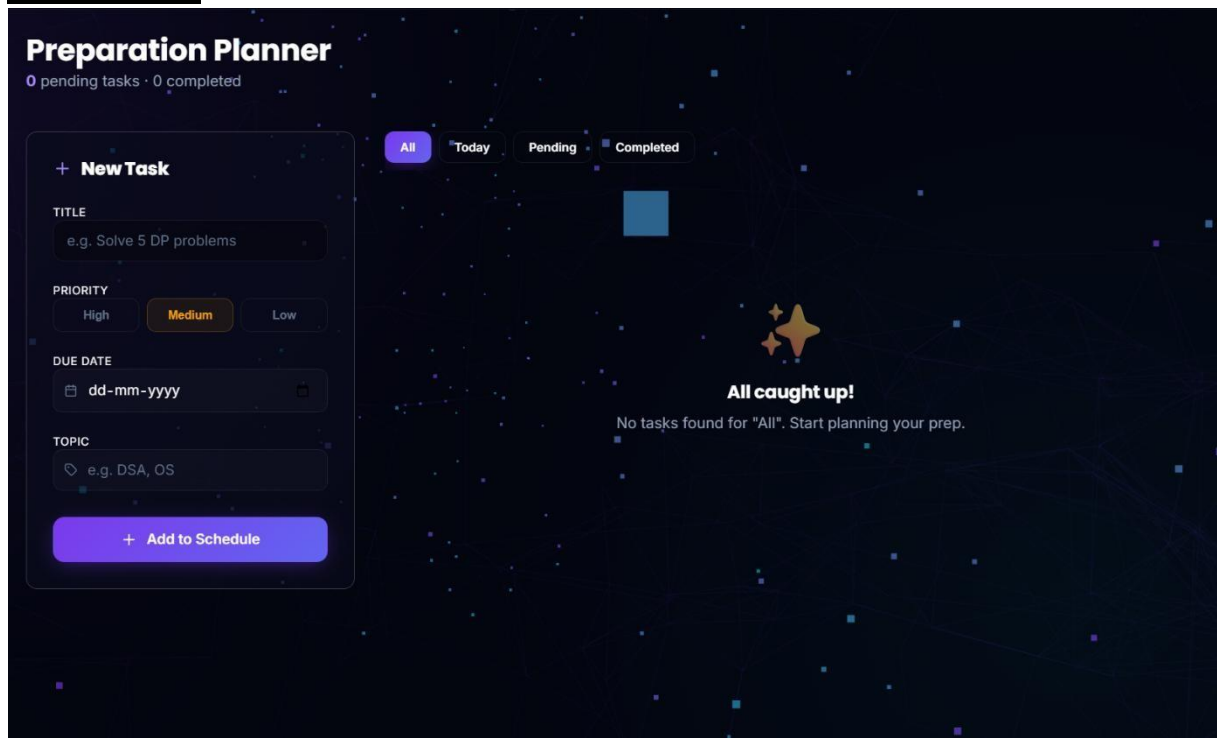
ELITE PROFESSIONAL VIEW

Your Full Name

< Previous Protocol

Next Protocol >

To-do planner



The image shows a 'Preparation Planner' interface with a dark, starry background. On the left is a 'New Task' form with fields for Title, Priority (High, Medium, Low), Due Date (dd-mm-yyyy), and Topic (e.g., DSA, OS), and an 'Add to Schedule' button. At the top right are filters for 'All', 'Today', 'Pending', and 'Completed'. The main area displays 'All caught up!' with a star icon and the message 'No tasks found for "All". Start planning your prep.'

Preparation Planner

0 pending tasks · 0 completed

+ New Task

TITLE
e.g. Solve 5 DP problems

PRIORITY
High Medium Low

DUE DATE
dd-mm-yyyy

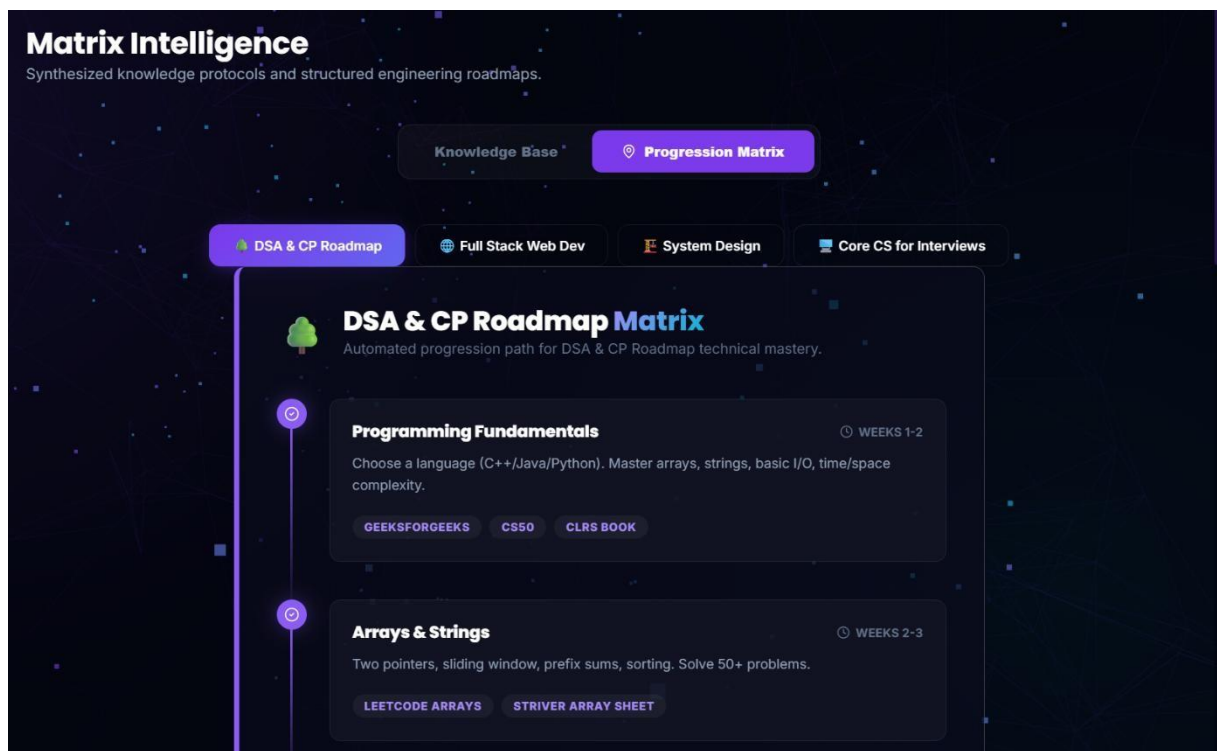
TOPIC
e.g. DSA, OS

+ Add to Schedule

All Today Pending Completed

All caught up!
No tasks found for "All". Start planning your prep.

Roadmap



Expected Outcome & Benefits

1. System Implementation

The PrepX platform was successfully designed, developed, and deployed as a full-stack web application. All planned modules were implemented and integrated into a single cohesive system. The application is accessible via any modern web browser and functions consistently across desktop, tablet, and mobile devices.

2. Module-wise Outcomes

- **User Authentication and Profile Management**

A secure, token-based authentication system was implemented using JSON Web Tokens (JWT) and bcryptjs password hashing. Users can register, log in, and maintain persistent sessions. Profile data including preparation streaks, topic scores, and completion statistics are stored and retrievable in real time.

- **Dashboard and AI Insights**

The personalized dashboard aggregates data from all modules — MCQ scores, mock interview performance, daily challenge streaks, and todo completion — into a single view. The AI Insights card, powered by the Anthropic API, analyzes user activity patterns and surfaces weak areas with targeted recommendations, providing a data-driven preparation roadmap.

- **MCQ Practice Module**

A curated bank of multiple-choice questions spanning Data Structures and Algorithms (DSA), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN), and Aptitude was implemented. Users receive instant feedback on each answer, and scores are tracked per topic to identify improvement areas over time.

- **Mock Interview Module**

The mock interview module presents timed, domain-specific questions that simulate real interview conditions. User responses are submitted to the AI evaluator (POST /api/ai/evaluate), which returns a score, qualitative feedback, and improvement suggestions. All sessions are persisted in MongoDB via the MockInterviewSession model for longitudinal performance tracking.

- **Resume Builder**

A step-by-step resume builder generates ATS-friendly resumes. Users input personal details, education, work experience, projects, and skills through a structured form. Data is saved against the user's account and can be updated at any time. The module reduces the time required to produce a professional resume from scratch.

- **Progress Analytics**

Visual progress reports display topic-wise scores, preparation streaks, and consistency metrics. The Progress model records each user's historical performance, enabling trend analysis across sessions. This feature directly supports self-regulated learning by making preparation gaps visible.

- **Daily Challenge and To-Do Planner**

Daily challenges provide a fresh question each day to maintain preparation momentum. The To-Do Planner allows users to create, manage, and mark tasks as complete, supporting structured study scheduling. Together these features promote daily engagement and habit formation.

- **Company-wise and Question Bank Modules**

A curated question bank allows filtering by topic and difficulty. The Company-wise module provides interview questions specific to commonly-targeted organizations, enabling targeted preparation aligned with placement goals.

Conclusion and Future Scope

Conclusion

PrepX was developed as a comprehensive, full-stack interview preparation platform to address a clearly identified gap: the fragmentation of preparation resources across multiple disconnected tools. Students typically rely on separate platforms for MCQ practice, mock interviews, resume building, and study planning — creating an inefficient and disjointed preparation experience.

This project demonstrates that a unified, AI-assisted platform can effectively consolidate these functions. Using React 18 on the frontend and a Node.js/Express REST API backed by

MongoDB on the server, PrepX delivers a seamless user experience with real-time progress tracking, AI-driven insights, and structured daily preparation workflows. The integration of the Anthropic API as an evaluation layer adds an intelligent dimension to practice sessions that static question banks cannot provide.

From a technical standpoint, the project successfully applied industry-standard practices including JWT authentication, HTTP security headers via Helmet.js, tiered rate limiting, structured logging with Winston, and a clean MVC-inspired layered architecture. These choices reflect a security-first mindset and ensure the platform is maintainable and extensible beyond the scope of this academic project.

The platform was validated across all planned modules: authentication, dashboard, MCQ practice, mock interviews, resume builder, progress analytics, daily challenges, company-wise preparation, notes, and to-do planning. All core features were implemented and integrated successfully, meeting the objectives set at the outset of the project.

In summary, PrepX fulfills its intended purpose as a centralized, intelligent, and responsive preparation companion for students approaching technical interviews and campus placement drives. It is a production-ready foundation that can be extended with additional features to serve a broader audience.

Future Scope

While the current version of PrepX delivers a complete feature set, several enhancements are planned for future iterations:

- **AI Voice Interview Assistant**
A real-time voice-based interview simulator using speech-to-text and text-to-speech technologies will allow users to practice verbal communication skills, a critical but currently unaddressed dimension of interview preparation. The existing AI evaluation pipeline can be extended to process spoken responses.
- **Personalized Study Roadmap Generation**
Based on a user's target company, role, and current skill level, the platform will generate a customized week-by-week preparation roadmap. This feature will leverage historical performance data from the Progress model to recommend the most impactful topics to study next.
- **Leaderboard and Gamification**
A competitive leaderboard displaying top performers by score, streak, and consistency will introduce healthy peer competition. Badges and achievement milestones will reward consistent engagement, further reinforcing the habit-building aspect of daily preparation.

- **Email Reminders and Push Notifications**

Automated daily reminders via email and browser push notifications will help users maintain their preparation streaks. Notification content will be personalized based on pending tasks and upcoming target interview dates.

- **Expanded Company-wise Preparation Paths**

The existing company-wise module will be extended into full preparation paths with multi-week syllabi, past interview experiences, and difficulty-graded question sets tailored to each organization's known interview patterns.

- **Collaborative Study Groups**

A peer study feature will allow users to form groups, share notes, challenge each other with custom MCQ sets, and conduct peer mock interviews. This social learning layer will complement the individual preparation experience.

- **Mobile Application**

A dedicated mobile application built with React Native will bring PrepX's full feature set to iOS and Android devices with native performance, offline access to question banks, and mobile-optimized UI components.

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